

Building Your First Profile Card

Units, Box Model & a Real Layout — Lab Worksheet

Name: _____ Section: _____ Date: _____

How to use this sheet: This is a lab session, so most of your time goes into **building** rather than writing. Move through the quick checks first — short answers only — then follow the **Main Task** step by step, checking your work in DevTools before moving on.

Part 1 — Units Warm-Up: px, %, vw, vh (Height & Width)

TRY THIS · VS CODE

Create a `div` with width of `300px`, height of `150px` and a `background-color`. Resize your browser window while watching the box.

```
div {  
  ...  
}
```

```
<body>  
  <div></div>  
</body>
```

Q1. Does the box move or change size as the window resizes? What does that confirm about `px`?

TRY THIS · VS CODE

Don't add any CSS for `<body>`. For the `div`, now make `width: 50%` & `height: 50%`.

Q2. Width shrinks and grows with the window as expected — but what happens to height? Any guess why `%` height behaves differently from `%` width here?

TRY THIS · VS CODE

Now, give `<body>` an explicit height: `400px`.

Q3. What changed? What does this tell you about what a `%` height actually needs from its parent?

TRY THIS · VS CODE

Create a new `<div>` with `width: 100vw; height: 100vh;`

Q4. Does this box fill the whole browser window no matter what size its parent is? How is that different from using `%`?

Part 2 — Exercise: Two Matching Boxes

TRY THIS · VS CODE

Build two `<div>` boxes with the exact same `width`, `height`, `padding`, and `margin`. Give them different `background-color` values — try red and green — so you can tell them apart, and put the text “Hello!” inside both boxes.

Q5. Now I want the red box to look bigger than the green box — but without changing its `height`, `width`, or `padding`. What can I add to make that happen? Try it, then explain what you changed and why it worked.

TRY THIS · VS CODE

Now stretch both boxes across the whole width of the page. Give both boxes `width: 100vw;`, `height: 250px;`, and `padding: 48px;`.

Q6. Look closely at your page. Do you see a horizontal scrollbar, or does part of a box get pushed off the screen? What do you think is causing this?

Q7. Based on what you just saw, which one should developers usually use for real projects — `content-box` (the default) or `border-box`? Why?

Part 3 — Main Task: Build a Profile Card

Build this step by step in a new file, `profile-card.html`. **Check each step in the browser before moving on to the next one** — don't wait until the end to look at it.

MAIN TASK · BUILD THE CARD (STEPS 1–5)

Step 1 — The Page Wrapper. Give the `<body>` (or a wrapper `<div>`) `width: 100vw; height: 100vh;` and a `background-color:` of your choice — this should fill the whole browser window, no matter what's inside it.

Step 2 — The Card. Inside the wrapper, add a `<div class="card">`. Give it `width: 400px;`, `background-color: white;`, `padding: 24px;`, `border: 1px solid #ddd;`, `box-sizing: border-box;`, and `margin: 40px auto;` to center it horizontally.

Step 3 — Profile Image Placeholder. Inside the card, add a `<div class="avatar">` with `width: 100px; height: 100px;`, a `background-color:` as a stand-in for a photo, and `margin: 0 auto 16px auto;` to center it and add space below it — the same centering trick from Part 2 of the last worksheet.

Step 4 — Name. Below the avatar, add an `<h2>` with a name. Give it `text-align: center;` and `margin: 0 0 8px 0;`.

Step 5 — Bio. Below the name, add a short `<p>` bio, one or two sentences. Give it `text-align: center;`, a muted `color:`, and `margin: 0;`.

MAIN TASK · STRETCH (OPTIONAL)

Change the card's `width: 400px;` to `width: 80%;` instead. Resize the browser from very narrow to very wide — does the card still look good, or does it break at either end?

Q8. Which box model properties did you use to build the card, and what did each one do?

Q9. You used three different units in this task — `vw/vh`, `px`, and maybe `%` in the stretch. Why did each unit make sense for the part of the page you used it on?
